# EXECUTIVE CREDIT RISK & PORTFOLIO INTELLIGENCE REPORT

Automated Underwriting, Macroeconomic Stress Testing, and ML Risk Scoring Engine

**Coverage:** 2.78M Loan Applications (CFPB HMDA) & Federal Reserve Macro Data (FRED) | **Date:** Sep 2026

## 1. EXECUTIVE SUMMARY

This institutional risk assessment evaluates a $1.11 Trillion retail credit portfolio. Integrating CFPB HMDA mortgage registers with Federal Reserve macroeconomic indicators, we deployed an automated underwriting and capital stress-testing framework to quantify credit vulnerability and institutional solvency.

| Total Pipeline Exposure | Overall Denial Rate | Model Champion AUC | Stress Loss Delta |
|---|---|---|---|
| **$1,108.54 Billion** | **25.44%** | **0.9973 (XGBoost)** | **2.18x (2008 Sc.)** |

**Strategic Highlights:**

- **Underwriting Velocity:** Production XGBoost engine enables sub-500ms automated approvals for qualified borrowers, capturing 70% straight-through processing (STP) volume.
- **Capital Protection:** Current credit policy effectively filtered 706,124 high-risk applications, averting $196.75 Billion in potential non-performing loans (NPL).
- **Capital Buffer Advisory:** Severe macro stress doubles Expected Losses to 44.38%, warranting a $24B-$48B counter-cyclical provisioning buffer.

## 2. PORTFOLIO & RISK PROFILE ANALYSIS

| State | Total Applications | Total Denied | Denial Rate | Total Exposure |
|---|---|---|---|---|
| California | 1,030,201 | 236,257 | 22.93% | $547.88 B |
| Texas | 854,669 | 217,244 | 25.42% | $277.43 B |
| Florida | 890,654 | 252,623 | 28.36% | $283.23 B |
| **Total Portfolio** | **2,775,524** | **706,124** | **25.44%** | **$1,108.54 B** |

- **Geographic Divergence:** Florida exhibits the highest rejection rate (28.36%) driven by high debt-to-income and property insurance friction. California anchors 49.4% of total exposure with the most resilient credit profile.
- **Ticket Size Vulnerability:** Micro-loans (<$100K) suffer ~50% rejection rates due to fragile debt service ratios, while core $250K-$500K loans maintain stable 22% denial rates.
- **Purpose Segmentation:** Home Purchase mortgages represent the safest credit asset (16.2% denial rate), whereas Home Improvement (45.2%) and Cash-Out Refinancing show heightened sensitivity.

## 3. MACROECONOMIC DRIVERS & MONETARY TRANSMISSION

| Effective Fed Funds | 30-Year Mortgage Rate | US Unemployment Rate | CPI Inflation (YoY) |
|---|---|---|---|
| 3.63% | 6.66% | 4.10% | 2.95% |

- **Rate Shock Impact:** 30Y mortgage rates hovering above 6.6% continue to restrict affordability, causing a 12-18 month lagging rise in broader banking delinquency rates (DRALACBS).
- **Labor Anchor:** Low unemployment (4.10%) provides critical debt-service solvency, preventing systemic foreclosure waves.

## 4. CAPITAL STRESS TESTING MATRIX (LGD = 45%)

| Macro Scenario | Mean PD (%) | Median PD (%) | PD > 50% Share | Expected Loss (EL) |
|---|---|---|---|---|
| Baseline (Current Economy) | 45.28% | 41.16% | 41.24% | 20.38% |
| Mild Recession | 70.16% | 71.02% | 85.85% | 31.57% |
| Severe Recession | 91.07% | 92.88% | 99.92% | 40.98% |
| 2008-like Housing Crisis | 98.61% | 99.09% | 99.98% | 44.38% |

- **Tail-Risk Non-Linearity:** Under severe shock conditions (Income -40%, Property Value -30%, DTI +60%), 99.98% of the loan book migrates into high-risk status, doubling credit loss exposure from 20.38% to 44.38%.

## 5. MACHINE LEARNING MODEL BENCHMARK & GOVERNANCE

| Model Architecture | AUC-ROC | Avg. Prec. | Precision | Recall | F1-Score | Role |
|---|---|---|---|---|---|---|
| Logistic Regression | 0.7550 | 0.5145 | 0.6520 | 0.7230 | 0.6857 | Baseline & Stress Test |
| XGBoost Classifier | 0.9973 | 0.9911 | 0.9940 | 0.9890 | 0.9915 | Production Scoring (STP) |

- **Dual Engine Strategy:** XGBoost powers real-time straight-through decisioning (<500ms latency), while Logistic Regression is maintained for regulatory compliance and macro sensitivity.
- **Model Monitoring:** Recalibration triggered if live AUC falls below 0.80 or Population Stability Index (PSI) exceeds 0.25.

## 6. STRATEGIC RISK COMMITTEE RECOMMENDATIONS

1. **Underwriting Optimization:** Implement automated tier-1 instant approvals for applicants with DTI < 36% and LTV < 80%.
2. **Geographic Risk Limit:** Cap portfolio concentration growth in Florida pending property insurance stabilization.
3. **Macro Capital Cushion:** Establish a $24B-$48B contingency capital buffer against stagflationary recession scenarios.
4. **Model Lifecycle Governance:** Enforce quarterly retraining and automated feature drift monitoring.